

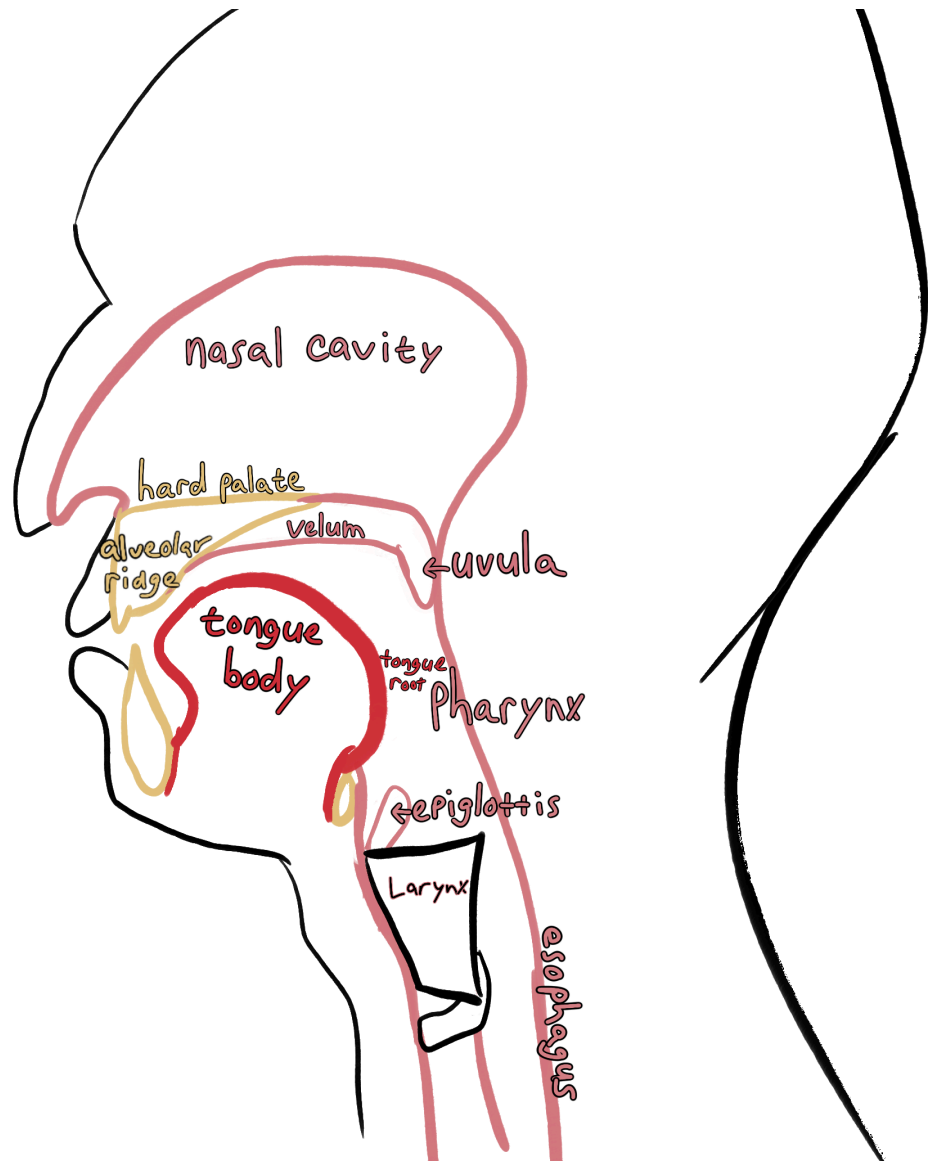


Speech Anatomy 4

Supralaryngeal Vocal Tract

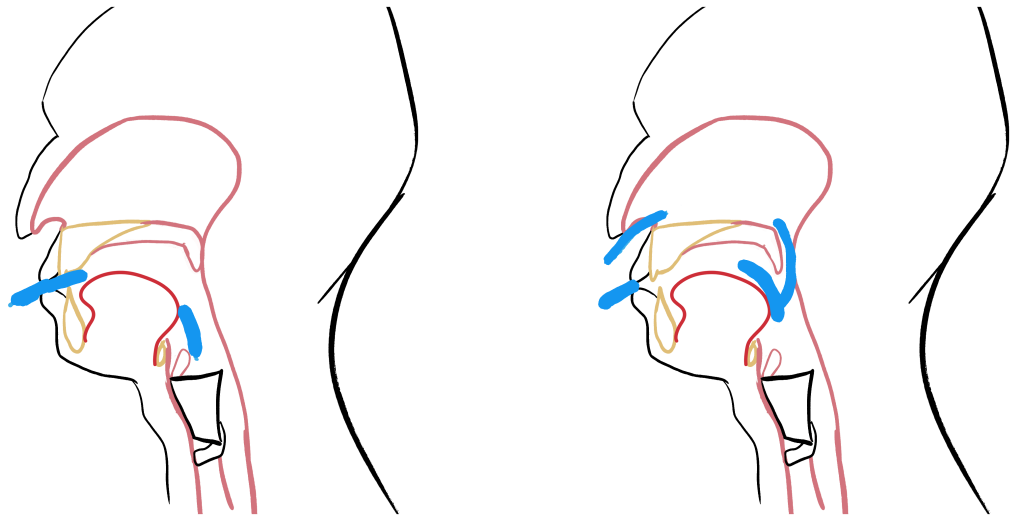
Dr. Josef Fruehwald

2026-09-01





Nasopharynx





The Tongue



The Tongue

- **extrinsic** muscles: muscles attached to the tongue
- **intrinsic** muscles: muscles within the tongue

The Tongue

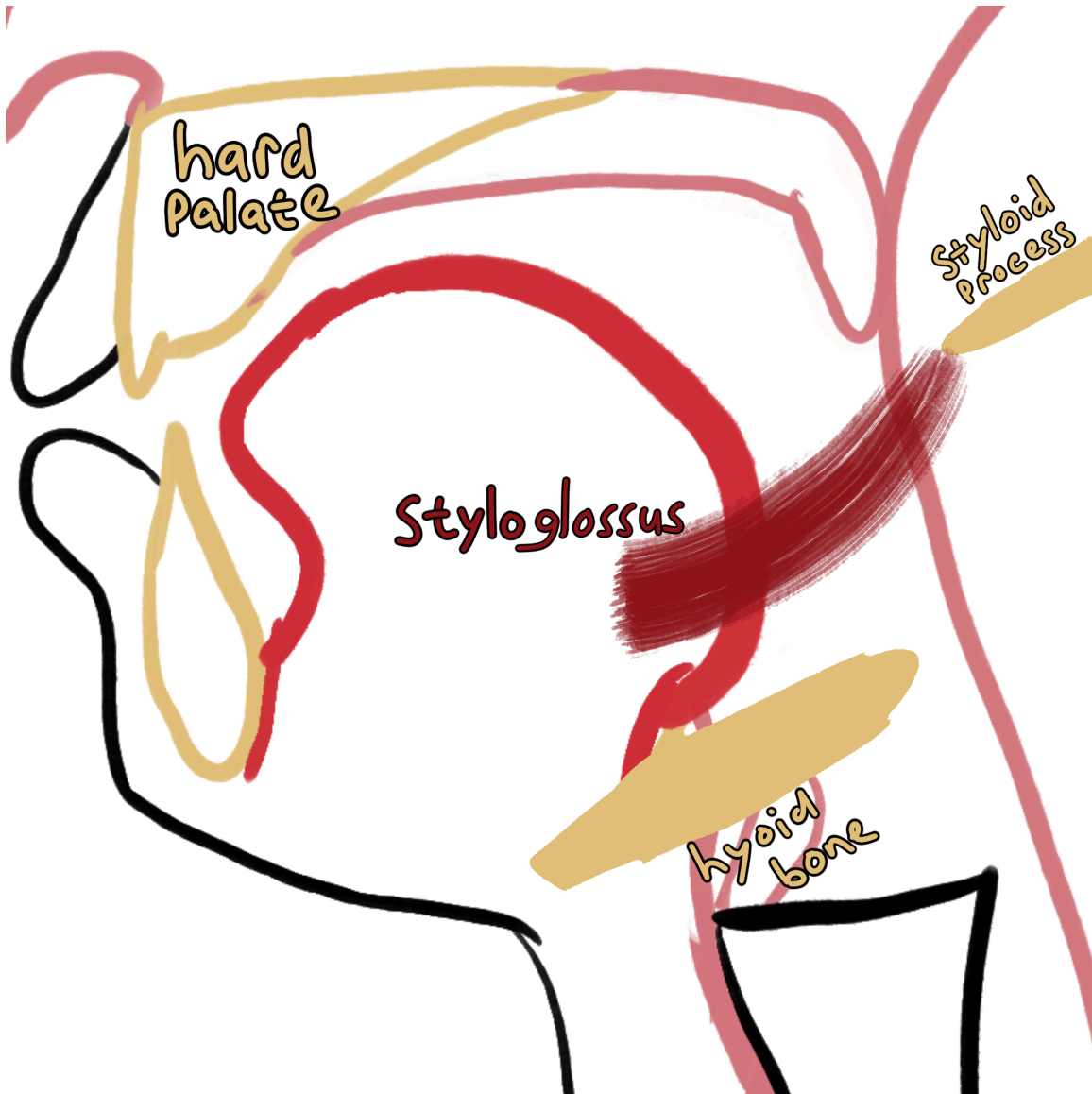
Muscles are usually named for the what they attach to.

- “-glossus”: Attaching to the tongue



The Tongue

Extrinsic Muscles



Pulls tongue body back



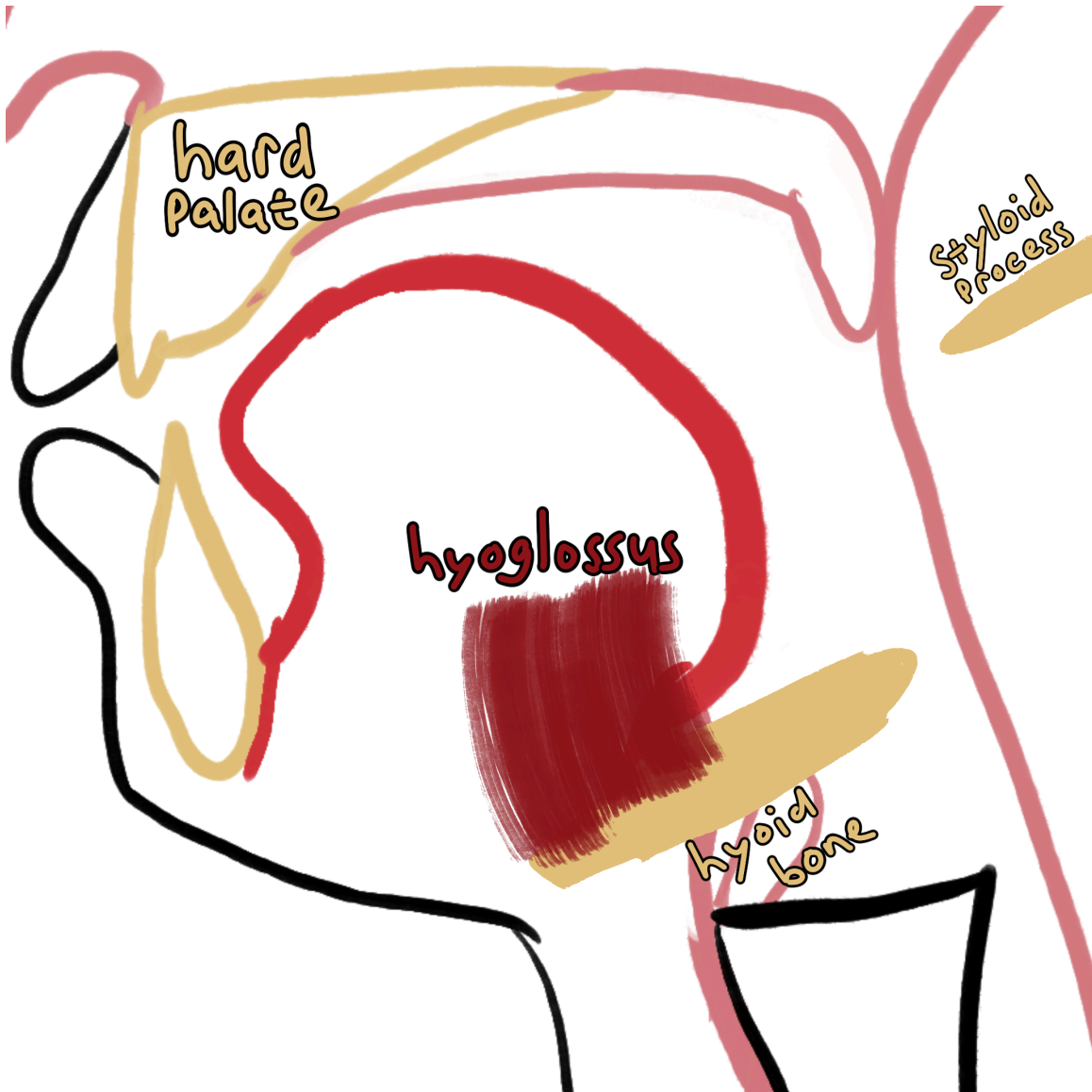
Styloid Process





The Tongue

Extrinsic Muscles

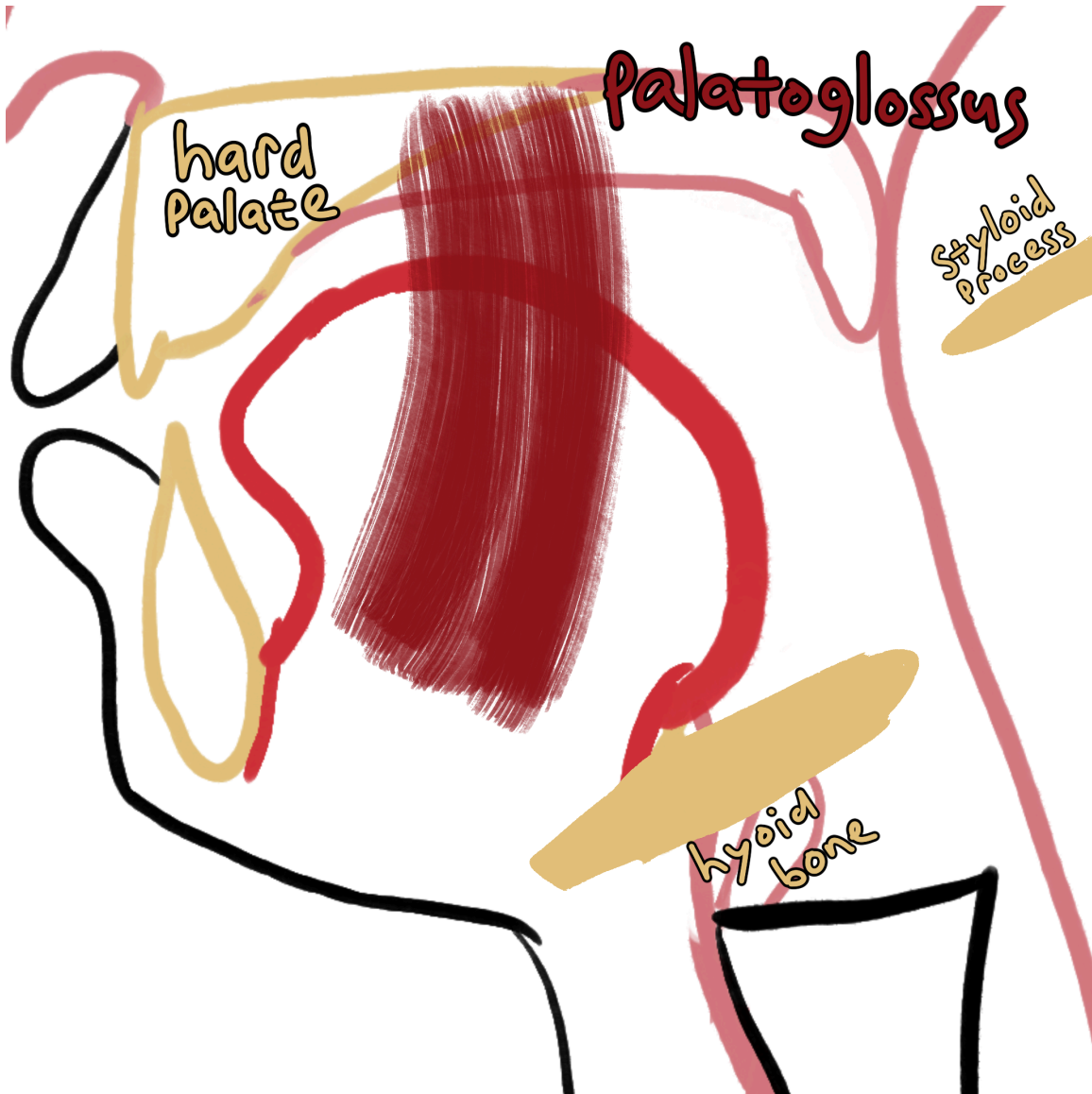


Pulls tongue body down



The Tongue

Extrinsic Muscles



Pulls tongue up



The Tongue

Extrinsic Muscles



Downward and forward

Coordinating it all

Everything is connected to everything

- Hyoglossus pulls down on tongue [?]
- Tongue pulls on palatoglossus [?]



- Palatoglossus pulls on soft palate

Low vowels are more likely to be nasalized than high vowels.

Coordinating it all

Timing

